VISION	MISSION
To be recognized for excellent engineering, developing global leaders both in educational and research in the domain of computer science and wireless engineering.	1. To create self-learning environment by facilitating leadership qualities, team spirit and ethical responsibilities. 2. To improve department-industry collaboration, interaction with professional society through technical knowledge and internship program. 3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering.

Experiment No. 05

Aim: - Write a program in Java to demonstrate following operations:

a) Single inheritance b) Multilevel inheritance

Software Requirement: - Java Development Kit (JDK), Integrated Development Environment (IDE) (NetBeans), Java Runtime Environment (JRE).

Theory: -

Inheritance

- Java, Inheritance is an important pillar of OOP (Object-Oriented Programming). It is the mechanism in Java by which one class is allowed to inherit the features (fields and methods) of another class.
- In Java, Inheritance means creating new classes based on existing ones.
- A class that inherits from another class can reuse the methods and fields of that class.
- In addition, you can add new fields and methods to your current class as well.

a) Single Inheritance

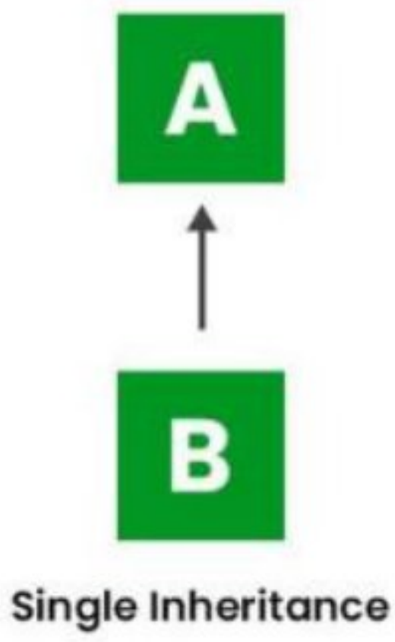
- In single inheritance, a sub-class is derived from only one super class.
- It inherits the properties and behaviour of a single-parent class. Sometimes, it is also known as simple inheritance.
- In the below figure, 'A' is a parent class and 'B' is a child class. The class 'B' inherits all the properties of the class 'A'.

Syntax:

```
// Superclass (Parent Class)
class SuperclassName {
    // Fields (Variables)
    // Methods
}
// Subclass (Child Class) inherits from Superclass
class SubclassName extends SuperclassName {
    // Additional Fields (Variables)
    // Additional Methods
```

Scanned with OKEN Scanner

VISION	MISSION
To be recognized for excellent engineering, developing global leaders both in educational and research in the domain of computer science and wireless engineering.	1. To create self-learning environment by facilitating leadership qualities, team spirit and ethical responsibilities. 2. To improve department-industry collaboration, interaction with professional society through technical knowledge and internship program. 3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering.



b) Multilevel Inheritance

- In Multilevel Inheritance, a derived class will be inheriting a base class, and as well as the derived class also acts as the base class for other classes.
- In the below image, class A serves as a base class for the derived class B, which in turn serves as a base class for the derived class C.
- In Java, a class cannot directly access the grandparent's members.

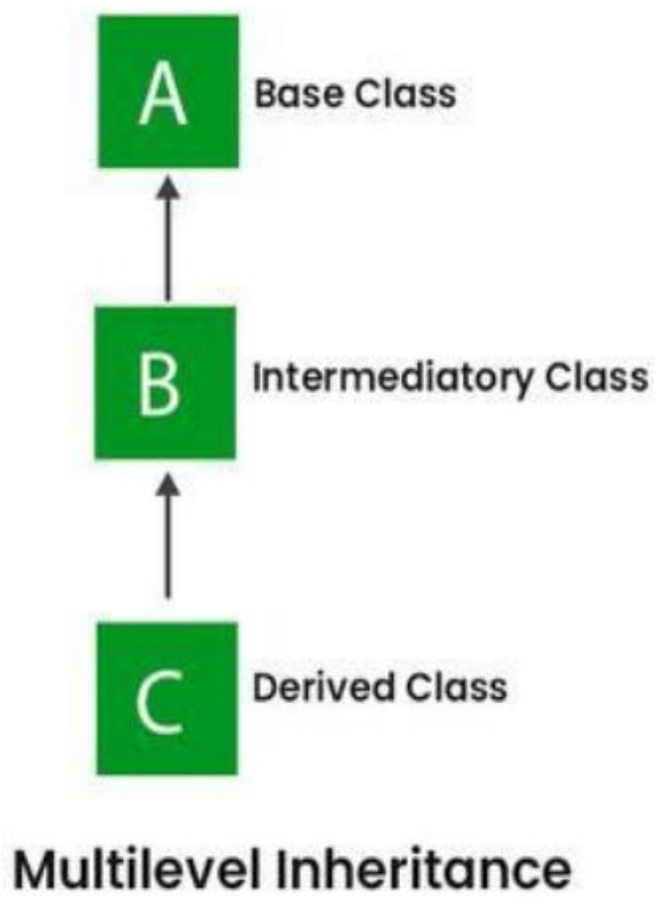
Syntax

```
// Base Class (Superclass)
class BaseClass {
   // Fields (Variables)
   // Methods
}

// Intermediate Class (Derived from Base Class)
class IntermediateClass extends BaseClass {
   // Additional Fields (Variables)
   // Additional Methods
}
// Derived Class (Derived from Intermediate Class)
class DerivedClass extends IntermediateClass {
   // Additional Fields (Variables)
```

Scanned with OKEN Scanner

```
    // Additional Methods
}
```



Program and Output

a) Single Inheritance

```
// Superclass
class Animal {
    void eat() {
        System.out.println("Animal is eating");
    }
}

// Subclass
class Dog extends Animal {
    void bark() {
        System.out.println("Dog is barking");
    }
```

Scanned with OKEN Scanner

```
}

// Main Class
public class Main {
    public static void main(String[] args) {
        // Create an object of the subclass
        Dog myDog = new Dog();

        // Access superclass method
        myDog.eat();

        // Access subclass method
        myDog.bark();
    }
}
```

Output:

```
Animal is eating
Dog is barking
```

b) Multilevel Inheritance

```
// Base Class
class Animal {
    void eat() {
        System.out.println("Animal is eating");
    }
}

// Intermediate Class
class Mammal extends Animal {
    void walk() {
        System.out.println("Mammal is walking");
    }
}
```

Scanned with OKEN Scanner

VISION	MISSION
To be recognized for excellent engineering, developing global leaders both in educational and research in the domain of computer science and wireless engineering.	1. To create self-learning environment by facilitating leadership qualities, team spirit and ethical responsibilities. 2. To improve department-industry collaboration, interaction with professional society through technical knowledge and internship program. 3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering.

```
// Derived Class
class Dog extends Mammal {
    void bark() {
        System.out.println("Dog is barking");
    }
}

// Main Class
public class Main {
    public static void main(String[] args) {
        // Create an object of the derived class
        Dog myDog = new Dog();

        // Access methods from Base Class
        myDog.eat();

        // Access methods from Intermediate Class
        myDog.walk();

        // Access methods from Derived Class
        myDog.bark();
    }
}
```

Output:

```
Animal is eating
Mammal is walking
Dog is barking
```

Conclusion: Hence, in this practical, we learned and executed the Single inheritance Multilevel inheritance in Java Successfully.

Scanned with OKEN Scanner